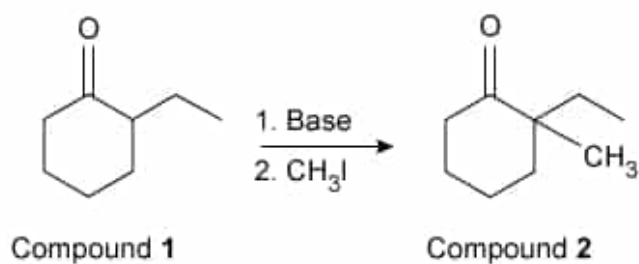


What base and reaction temperature are needed to add a methyl group to Compound **1** as shown in the reaction below?

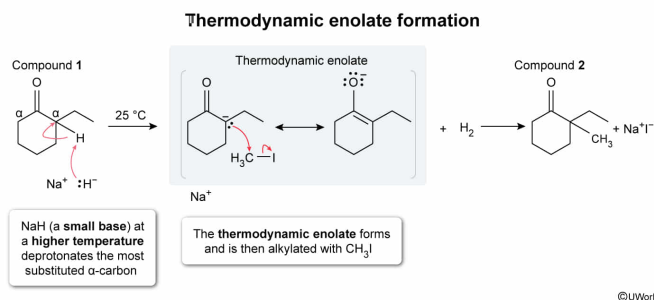


- ☐ A. LDA; $-78\text{ }^{\circ}\text{C}$ [15%]
- ☐ B. LDA; $25\text{ }^{\circ}\text{C}$ [37%]
- ☐ C. NaH; $-78\text{ }^{\circ}\text{C}$ [10%]
- ✓ ☒ D. NaH; $25\text{ }^{\circ}\text{C}$ [35%]

Correct

35% answered correctly

Explanation:



An **enolate** is formed when a base **deprotonates** an α -carbon (ie, removes an acidic proton from the carbon adjacent to a carbonyl $\text{C}=\text{O}$). For unsymmetrical carbonyl compounds, the type of base used and the reaction temperature affect which α -carbon is deprotonated. The deprotonated α -carbon can then be methylated by reacting with a methyl halide, such as CH_3I .

The **kinetic enolate** results from deprotonation of the **least substituted α -carbon**. This enolate forms relatively quickly because the base can access the less substituted carbon more easily. Consequently, a **large, bulky base**, such as **lithium diisopropylamide** (LDA), and a lower reaction temperature (eg, $-78\text{ }^{\circ}\text{C}$) may be used to yield this product. Alternatively, the **thermodynamic enolate** forms by deprotonation of the **more substituted α -carbon**. A **small base** such as sodium hydride (NaH) and a higher reaction temperature (eg,

25 °C) are needed to facilitate this reaction. The thermodynamic enolate is more stable but forms more slowly than the kinetic enolate.

Compound **1** in this question is an unsymmetrical ketone in which one α -carbon has one substituent (the 6-membered ring) and the other has two substituents (the 6-membered ring and an ethyl group). The reaction shows Compound **1** becomes methylated at the more substituted α -carbon. For this to occur, the thermodynamic enolate must be created using a small base and a higher reaction temperature. Of the choices given, NaH and 25 °C are ideal for thermodynamic enolate formation.

(Choices A and B) Because of **steric hindrance**, LDA is too bulky to deprotonate the more substituted α -carbon and is used to form the **kinetic** rather than the thermodynamic enolate.

(Choice C) Thermodynamic enolates are formed at *higher* reaction temperatures. -78 °C is too low for the formation of the thermodynamic product.

Educational objective:

Kinetic enolates are formed by deprotonation of the least substituted α -carbon of a carbonyl compound using a bulky base, such as LDA at low temperatures. Thermodynamic enolates are formed by deprotonation of the more substituted α -carbon using a small base, such as NaH, at higher temperatures.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403333

System:
Functional Groups and Their Reactions